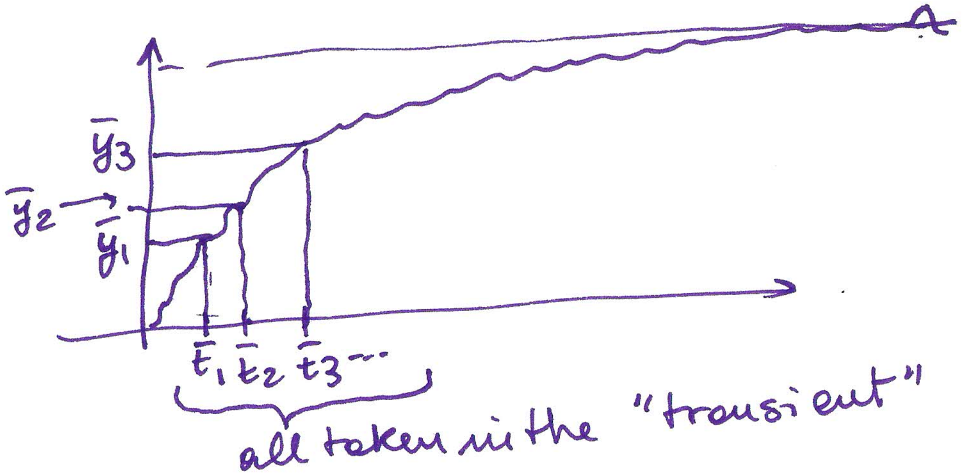
3rd method also suffers of num. p.bs.

Possibilities (for every method):

make a number of experiments, then average on the results.

if the measurement noise is zero-mean $\Rightarrow$ ok

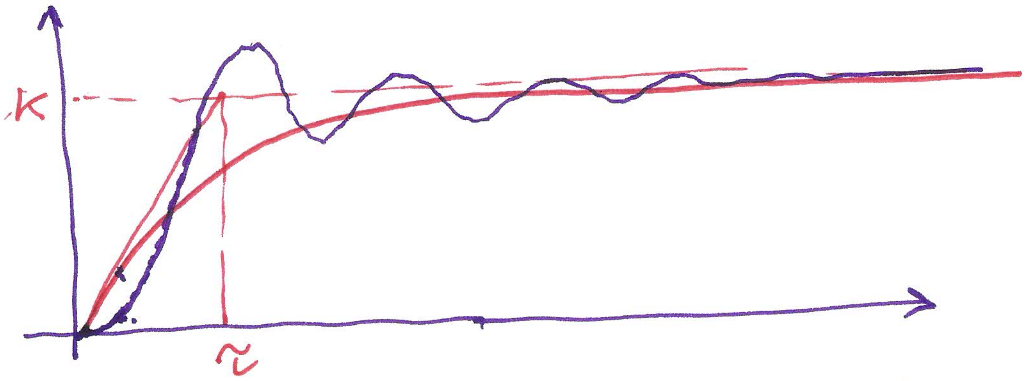
Another possibility for method 3 consist in using more then one values for $\bar{t}$ $\Rightarrow$ $\bar{t}_1, \bar{t}_2, \ldots$



As a general idea: we can use many values over time and many realizations of the esperiment.

What happens if we use our Taylor-made method for 1st order system on a system that does not obey to a first order model?

ex

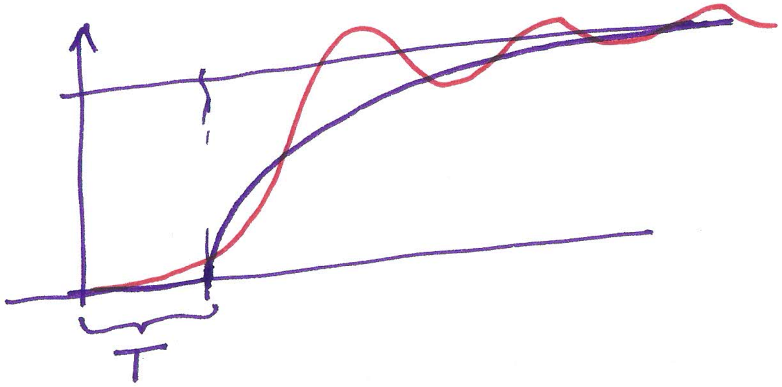


Using a 1st order model is not "wrong" but "not so good".
Red line ≡ 1st order model "Best possible" (?) 1st order model will be a "poor" model, but maybe "not so bad."

Attention: if we apply ~~1st~~ method (1) for 1st ord systems here
⇒ we obtain τ very big. (tangent in 0 is almost 0)

⇒ method 1 is not good.

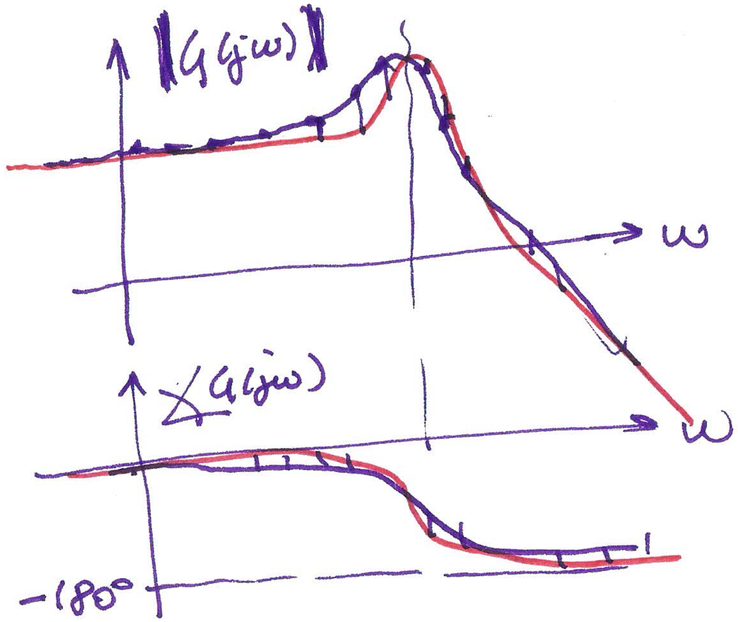
One possibility is to include a delay



Note on relations between grey box models and "Bode Plots".

If we obtain a Bode Plot experimentally (see I13) ~~maybe~~ in principle we do not have a mathem. model, BUT we can use the data on the Bode Plot to try to fit some mathematical model.

Ex



$\Rightarrow$ a good approx could be a second order transf. function.

$$G(s) = \frac{K}{1 + \frac{2\xi}{\omega_0}s + \frac{s^2}{\omega_0^2}}$$

Only 3 parameters

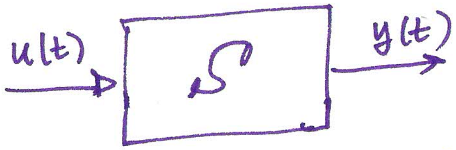
K, ξ, ω_0

$\Rightarrow$ identification by means of (for ex.) least squares method.

If a 2nd order system is not sufficiently good (validation!) $\Rightarrow$ we can use more parameters.
(one zero, or more poles, or more zeros & poles).

(Some) introduction to models for linear time invariant systems

Continuous-time



t for "cont. time".

Consider S.I.S.O. (single In single Out)

- differential eq.s.
- state eq.s
- Transf. function (impulse response)
  (only represents controllable & observ. part of the system)

$$y(t) = y_f(t) + y_\phi(t)$$

$y_f(t)$: forced output, y when initial cond are 0 (depend only on ~~u(t)~~ $u(\cdot)$)

$y_\phi(t)$: free response, y when $u(\cdot)$ is zero, depends only on initial conditions.

$y_\phi(t) \xrightarrow[t\to\infty]{} 0$ for asymptotically stable systems (or, at least, when such is the observ. part) if $u(t) = \delta(t) \Rightarrow y_f(t) = g_c(t)$ IMPULSE RESP.

in general

$$u(t) \Longrightarrow \boxed{y_f(t) = g_c(t) * u(t)}$$

cont.t. convolution $\int_0^t g_c(\tau)\, u(t-\tau)\, d\tau$

Transfer function:

$$G_c(s) = \mathcal{L}\{g_c(t)\} = \int_{0^-}^{+\infty} g_c(t)\, e^{-st}\, dt$$

Laplace transf.

$$\boxed{Y_f(s) = G_c(s) \cdot U(s)}$$

$$Y_f(s) = \mathcal{L}\{y_f(t)\} \qquad U(s) = \mathcal{L}\{u(t)\}$$

Discrete time systems

$$u(k) \longrightarrow \boxed{S} \longrightarrow y(k)$$

Abuse of notation.

$$u(k) \equiv u(t)\Big|_{t = k\Delta T} \qquad y(k) = y(k\Delta T)$$

ΔT: sample time

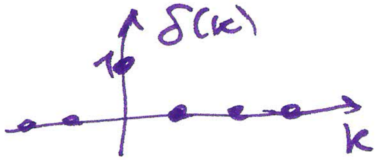
- difference equations
- state equations.
- Transf. function / impulse resp.

$$y(k) = y_f(k) + y_0(k)$$

$y_0(k) \xrightarrow[k\to\infty]{} 0$ for asympt. stable syst. (...)

if $u(k) = \delta(k) = \begin{cases} 1 & k=0 \\ 0 & k\neq 0 \end{cases}$

$\uparrow$ discrete time impulse



$\Rightarrow y_f(k) = g(k)$ IMPULSE RESP.

in general

$$y_f(k) = g(k) * u(k) = \sum_{\ell=0}^{\infty} g(\ell)\,u(k-\ell)$$

$\uparrow$ discrete convolution

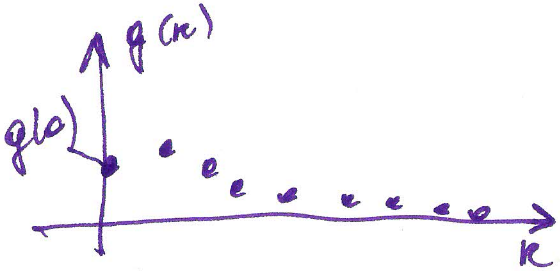
$$= g(0)u(k) + g(1)\,u(k-1) + g(2)u(k-2) + \dots$$

often $g(0)$ is assumed to be zero (e.g. Young)

$\rightarrow \sum_{\ell=1}^{\infty} g(\ell)\,u(k-\ell)$

$\uparrow$ if $g(0)=0$ [strictly proper systems]

We do not assume $g(0)=0$



We could use the $\mathcal{Z}$-transform

for a seq $\{v(k), k=0,1,\dots\}$

$$V(z) = \mathcal{Z}\{v(k)\} = \sum_{k=0}^{\infty} v(k) z^{-k}$$

we have

$$G(z) = \mathcal{Z}\{\underbrace{g(k)}_{\text{imp. resp.}}\} = \sum_{k=0}^{\infty} g(k) z^{-k}$$

$$= g(0) + g(1) z^{-1} + g(2) z^{-2} + \dots$$

and

$$\underbrace{Y_f(z)}_{\mathcal{Z}\{y_f(k)\}} = G(z) \underbrace{U(z)}_{\mathcal{Z}\{u(k)\}}$$

Hybrid Notation

We interpret "z^{-1}" as the "one step delay" **operator**

Given a sequence $\{v(k)\} \Rightarrow$

$$z^{-1} v(k) = v(k-1)$$

$$z^{-m} v(k) = v(k-m)$$

Re-consider

$$y_f(k) = \sum_{\ell=0}^{\infty} g(\ell)\, u(k-\ell) = g(0)u(k) + g(1)u(k-1) + g(2)u(k-2) \ldots$$

consider also

$$G(z) = \sum_{\ell=0}^{\infty} g(\ell)\, z^{-\ell} = g(0) + g(1)z^{-1} + g(2)z^{-2}$$

If we write $G(z) \cdot u(k) = \underbrace{\left[\sum_{\ell=0}^{\infty} g(\ell) z^{-\ell}\right]}_{G(z)} u(k) =$

$$= \sum_{\ell=0}^{\infty} g(\ell) \underbrace{z^{-\ell} u(k)}_{= u(k-\ell)} = \sum_{\ell=0}^{\infty} g(\ell)\, u(k-\ell)$$

$$= y_f(k)$$

(the same) without sums:

$$G(z)u(k) = \underbrace{\left[g(0) + g(1)z^{-1} + g(2)z^{-2} + \ldots\right]}_{G(z)} u(k) =$$

$$= g(0)u(k) + g(1)z^{-1}u(k) + g(2)z^{-2}u(k) + \ldots =$$

$$= g(0)u(k) + g(1)u(k-1) + g(2)u(k-2) + \ldots$$

We have obtained $\boxed{y_f(k) = G(z)\, u(k)}$

z, k Hybr. notation

"z^{-1}" is the one step delay

$G(z)$ is exactly the transf. function.

$\Rightarrow$ $G(z)$ is interpreted as an operator.

<u>N.B.</u> $G(z) = g(0) + g(1)z^{-1} + g(2)z^{-2} + \dots$

where $g(k)$, $k = 0, 1, 2, \dots$ are the values of the impulse resp. of the system.

<u>Ex.</u> $G(z) = 1$

$$u(k) \longrightarrow \boxed{G(z)=1} \longrightarrow y_f(k) = u(k)$$

$\Rightarrow$ $y_f(k) = G(z)\,u(k) = u(k)$

NOTE $G(z) = 1.$ $g(k) = ?$ $\therefore$ $\begin{cases} 1 & k=0 \\ 0 & k>0 \end{cases}$ $\Rightarrow$ $g(k) = \delta(k)$

$$G(z) = \overbrace{\underbrace{1}_{g(0)} + \underbrace{0}_{g(1)} \cdot z^{-1} + \underbrace{0}_{g(2)}\, z^{-2} + \underbrace{0}_{g(3)}\, z^{-3} + 0 \dots}$$

$G(z) = z^{-1}$

$$y_f(k) = G(z)\,u(k) = \underbrace{z^{-1}}_{=\frac{1}{z}}\,u(k) = u(k-1)$$